IMPORTANT INSTRUCTIONS

Please ensure below pointers are met while submitting the Idea PPT:

1. **Kindly keep the maximum slides limit up to six (6). (Including the title slide)**
2. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
3. **Keep your explanation precise and easy to understand**
4. **Idea should be unique and novel.**
5. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**
6. **You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.**

Note - You can delete this slide (Important Pointers) when you upload the details of your idea portal.

TITLE PAGE

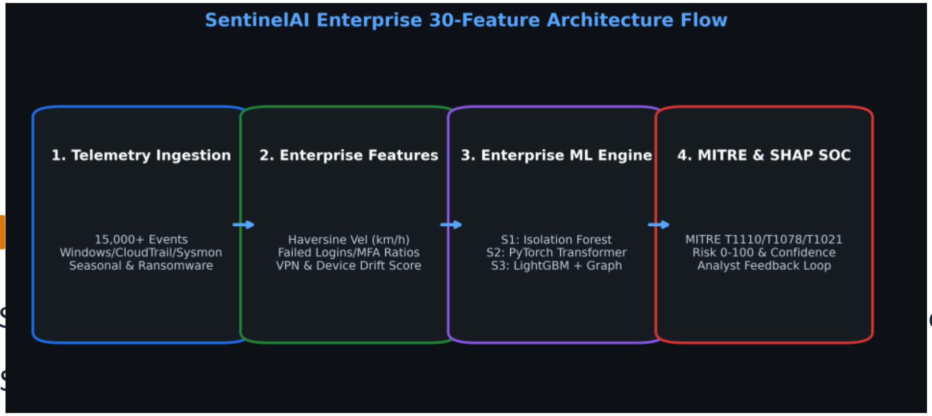
- **Problem Statement ID –**
- **Problem Statement Title-**
- **Theme-**
- **PS Category- Software/Hardware**
- **Student Name (Registered on portal)**
- **Student ID**

IDEA TITLE

PROPOSED SOLUTION & INNOVATION

- **Idea Title:** SentinelAI — AI-Powered Cascaded Anomaly & Threat Detection Engine
- **Core Solution:** Sequence-aware ML engine detecting zero-day lateral movement, stolen credential abuse, brute force, & ransomware.
- **Multi-Entity Telemetry Delineation:** Explicit operational delineation across Users (👤), Service Accounts (⚙️), and Edge Devices (🖥️).
- **Auth Method Tracking:** Monitors 4 distinct authentication protocols: Password, Token (OAuth/SAML/JWT), Certificate (mTLS), and Biometric (Passkey/FIDO2).
- **Explainability & MITRE Mapping:** Local SHAP feature attributions paired with MITRE ATT&CK technique IDs (T1078, T1110, T1041, T1486) for plain-English threat storytelling.
- **Active SOC Containment:** 1-click response buttons for session token revocation, credential reset, and network host endpoint isolation.

TECHNICAL APPROACH



• TECHNICAL APPROACH & ARCHI

• **Core ML & Backend:** Python 3.12, FastAPI (REST & WebS … e.

• **Frontend SOC Dashboard:** React 18, Vite 5, Tailwind CSS …

• **Feature Engineering Pipeline:** Extracts 30 spatiotemporal & sequence features including Haversine Velocity (km/h), cyclical time, and device drift.

• **Cascaded 3-Stage Architecture:** Stage 1 Unsupervised Anomaly Scoring -> Stage 2 Sequence Window -> Stage 3 LightGBM Multi-Class Classifier.

• **Model Disk Caching:** Pre-trained models serialized with joblib for 0.06s instant server startup and sub-millisecond inference per log event.

FEASIBILITY AND VIABILITY

• FEASIBILITY, VIABILITY & RISK MITIGATION

• **Feasibility & Performance:** 0.06s server launch time, ~1,200 events/sec throughput, and clean production Vite build (10.72s compile time).

• **Analyst Alert Budget:** Dynamic 97th percentile cutoff isolates top 1% critical threats, reducing alert fatigue by 99%.

• **PSI Concept Drift Monitoring:** Tracks population stability index (PSI = 0.0167) to automatically detect data drift and maintain model accuracy.

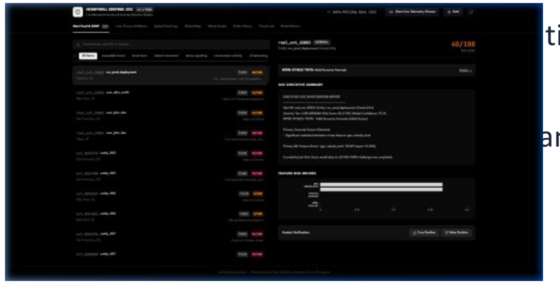
• **Potential Challenges & Risks:** High-velocity telemetry bursts during DDoS/brute-force attacks and risk of false positives interrupting business workflows.

• **Mitigation Strategies:** Analyst Feedback Loop (/api/feedback) for online active learning + 1-click SOC false-positive dismissal.

ARTIFACTS

- **SYSTEM ARTIFACTS & LIVE DASHBOARD PROTOTYPE**

• **Modular Codebase:** backend/api_server.py (FastAPI core), backend/feature_engine.py, frontend/src/components/UserProcessDashboard.jsx.



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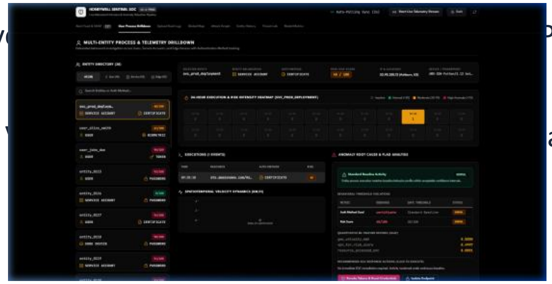

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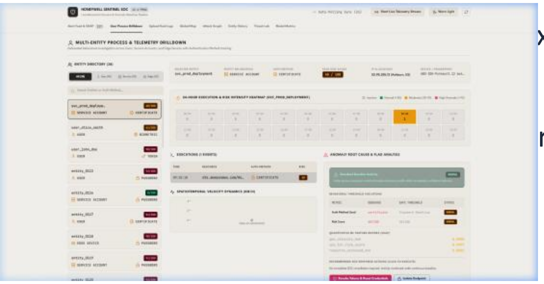

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RESEARCH AND REFERENCES

• RESEARCH, FRAMEWORKS & REFERENCES

• **MITRE ATT&CK Framework:** Mapped detection logic to T1078 (Valid Accounts), T1110 (Brute Force), T1041 (Exfiltration), & T1486 (Ransomware Encrypt).

• **SHAP Explainability (Lundberg & Lee, 2017):** A Unified Approach to Interpreting Model Predictions (NeurIPS 2017) for quantitative feature attributions.

• **Haversine Spatiotemporal Dynamics:** Great-Circle Distance calculation for high-velocity physical travel anomaly detection (>800 km/h).

• **LightGBM Engine (Ke et al., 2017):** LightGBM: A Highly Efficient Gradient Boosting Decision Tree (Advances in Neural Information Processing Systems 30).

• **Project Documentation & Repository:** Complete technical documentation in README.md and PDF Guide (AI_Powered_Behavioral_Anomaly_Detection_Comprehensive_Guide.pdf).